

ASME B36.19M-2018
[Revision of ASME B36.19M-2004 (R2015)]

Stainless Steel Pipe

AN AMERICAN NATIONAL STANDARD



**The American Society of
Mechanical Engineers**

ASME B36.19M-2018
[Revision of ASME B36.19M-2004 (R2015)]

Stainless Steel Pipe

AN AMERICAN NATIONAL STANDARD



**The American Society of
Mechanical Engineers**

Two Park Avenue • New York, NY • 10016 USA

Date of Issuance: September 19, 2018

This Standard will be revised when the Society approves the issuance of a new edition.

ASME issues written replies to inquiries concerning interpretations of technical aspects of this Standard. Interpretations are published on the Committee web page and under <http://go.asme.org/InterpsDatabase>. Periodically certain actions of the ASME B32 Committee may be published as Cases. Cases are published on the ASME website under the B32 Committee Page at <http://go.asme.org/B32committee> as they are issued.

Errata to codes and standards may be posted on the ASME website under the Committee Pages to provide corrections to incorrectly published items, or to correct typographical or grammatical errors in codes and standards. Such errata shall be used on the date posted.

The B32 Committee Page can be found at <http://go.asme.org/B32committee>. There is an option available to automatically receive an e-mail notification when errata are posted to a particular code or standard. This option can be found on the appropriate Committee Page after selecting “Errata” in the “Publication Information” section.

ASME is the registered trademark of The American Society of Mechanical Engineers.

This code or standard was developed under procedures accredited as meeting the criteria for American National Standards. The Standards Committee that approved the code or standard was balanced to assure that individuals from competent and concerned interests have had an opportunity to participate. The proposed code or standard was made available for public review and comment that provides an opportunity for additional public input from industry, academia, regulatory agencies, and the public-at-large.

ASME does not “approve,” “rate,” or “endorse” any item, construction, proprietary device, or activity.

ASME does not take any position with respect to the validity of any patent rights asserted in connection with any items mentioned in this document, and does not undertake to insure anyone utilizing a standard against liability for infringement of any applicable letters patent, nor assume any such liability. Users of a code or standard are expressly advised that determination of the validity of any such patent rights, and the risk of infringement of such rights, is entirely their own responsibility.

Participation by federal agency representative(s) or person(s) affiliated with industry is not to be interpreted as government or industry endorsement of this code or standard.

ASME accepts responsibility for only those interpretations of this document issued in accordance with the established ASME procedures and policies, which precludes the issuance of interpretations by individuals.

No part of this document may be reproduced in any form,
in an electronic retrieval system or otherwise,
without the prior written permission of the publisher.

The American Society of Mechanical Engineers
Two Park Avenue, New York, NY 10016-5990

Copyright © 2018 by
THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS
All rights reserved
Printed in U.S.A.

CONTENTS

Foreword	iv
Committee Roster	v
Correspondence With the B32 Committee	vi
1 Scope	1
2 Size	1
3 Materials	1
4 Wall Thickness	1
5 Weights/Masses	1
6 Permissible Variations	1
7 Pipe Threads	1
8 Wall-Thickness Selection	2
Table	
2-1 Dimensions of Welded and Seamless Stainless Steel Pipe and Nominal Weights (Masses) of Steel Pipe, Plain End	3

FOREWORD

This Standard for corrosion-resistant piping, designated categorically as *stainless*, is based on the same principles that formed the background for the development of ASME B36.10M, Welded and Seamless Wrought Steel Pipe, and reference is made to this source of information.

The more recent development of the highly alloyed stainless steels has brought about a minor conflict with convention. With these newer materials, the need for standards is just as great and the present types of threads are just as satisfactory, but the basic cost of the metal is much higher and the art of fusion welding has developed concurrently. The character of stainless steel permits the design of thin-wall piping systems without fear of early failure due to corrosion, and the use of fusion welding to join such piping has eliminated the necessity of threading it. For these reasons, the wall-thickness dimensions shown under Schedule 10S have been developed, based on the conventional formula, but then modified to correspond to the nearest Birmingham Wire Gage (B.W.G.) number.

Following publication of the 1949 edition, a demand developed for a still lighter wall pipe. A Schedule 5S was determined cooperatively by representatives of chemical companies, processing industries, and manufacturers of welding fittings. This was endorsed by the American Standards Association (ASA) Chemical Industry Correlating Committee and the Manufacturers Standardization Society of the Valve and Fittings Industry. The new schedule was included in the revised standard that was approved by ASA [now American National Standards Institute (ANSI)] on April 7, 1952.

In 1956, it was recommended that the wall thickness of 12 in. 5S be lessened, and a new revision of the standard was issued shortly after its approval by ASA on February 27, 1957. In this fourth edition, dimensions were expanded beyond 12 in. pipe size by inclusion of, and reference to, ASTM Specification A409. This revision was approved by ASA on October 29, 1965.

The B36 Standards Committee membership was asked in March 1970 for recommendations as to what action should be taken on ANSI B36.19-1965 since, according to ANSI procedures, this Standard was due for revision or affirmation. The B36 Standards Committee recommended reaffirmation. This action was approved by the Secretariat and by ANSI on May 26, 1971.

In 1975, the B36 Standards Committee undertook a review of the standard, considering its acceptability and usefulness. The results were favorable; some editorial refinements and updating were proposed, along with the incorporation of factors for conversion to SI (metric) units. The revision was approved by the Standards Committee, the Secretariat, and subsequently ANSI on October 4, 1976.

The standard was revised in 1984 to include SI (metric) dimensions. The outside diameters and wall thicknesses were converted to millimeters by multiplying the inch dimensions by 25.4. Outside diameters larger than 16 in. were rounded to the nearest 1 mm, and outside diameters 16 in. and smaller were rounded to the nearest 0.1 mm. Wall thicknesses were rounded to the nearest 0.01 mm. These converted and rounded SI dimensions were added. A formula to calculate the SI plain end mass, kg/m, using SI diameters and thicknesses, was added, and the calculations were added. These changes in the standard were approved by the Standards Committee, the Sponsor, and ANSI, and it was designated an American National Standard on October 7, 1985.

The text of the standard was revised in the 2004 edition to conform to the format and content, as appropriate, of ASME B36.10M-2004. A new table was added, combining the information in the previous tables into a single table. Also, the roster of the disbanded B36 Committee was replaced by the roster of the B32 Committee. The 2004 edition was approved as an American National Standard on June 23, 2004.

The 2018 edition revises some of the outside diameters and plain end masses in [Table 2-1](#) (formerly Table 1). A number of editorial revisions have also been made to [Table 2-1](#) and [sections 1, 2, 4, 5, and 7](#). The 2018 edition was approved as an American National Standard on August 13, 2018.

ASME B32 COMMITTEE

Metal and Metal Alloy

Wrought Mill Product Nominal Sizes

(The following is the roster of the Committee at the time of approval of this Standard.)

STANDARDS COMMITTEE OFFICERS

F. W. Tatar, *Chair*
A. R. Amaral, *Secretary*

STANDARDS COMMITTEE PERSONNEL

A. R. Amaral, The American Society of Mechanical Engineers
D. O. Bankston, Jr., Bechtel
D. Frikken, Becht Engineering Co.
J. A. Gruber, J A Gruber & Associates
R. A. McLeod, General Electric Co.
M. L. Nayyar, NICE
A. P. Rangus, Bechtel
R. Reamy, Turner Industries Group
F. W. Tatar, FM Global

CORRESPONDENCE WITH THE B32 COMMITTEE

General. ASME Standards are developed and maintained with the intent to represent the consensus of concerned interests. As such, users of this Standard may interact with the Committee by requesting interpretations, proposing revisions or a case, and attending Committee meetings. Correspondence should be addressed to:

Secretary, B32 Standards Committee
The American Society of Mechanical Engineers
Two Park Avenue
New York, NY 10016-5990
<http://go.asme.org/Inquiry>

Proposing Revisions. Revisions are made periodically to the Standard to incorporate changes that appear necessary or desirable, as demonstrated by the experience gained from the application of the Standard. Approved revisions will be published periodically.

The Committee welcomes proposals for revisions to this Standard. Such proposals should be as specific as possible, citing the paragraph number(s), the proposed wording, and a detailed description of the reasons for the proposal, including any pertinent documentation.

Proposing a Case. Cases may be issued to provide alternative rules when justified, to permit early implementation of an approved revision when the need is urgent, or to provide rules not covered by existing provisions. Cases are effective immediately upon ASME approval and shall be posted on the ASME Committee web page.

Requests for Cases shall provide a Statement of Need and Background Information. The request should identify the Standard and the paragraph, figure, or table number(s), and be written as a Question and Reply in the same format as existing Cases. Requests for Cases should also indicate the applicable edition(s) of the Standard to which the proposed Case applies.

Interpretations. Upon request, the B32 Standards Committee will render an interpretation of any requirement of the Standard. Interpretations can only be rendered in response to a written request sent to the Secretary of the B32 Standards Committee.

Requests for interpretation should preferably be submitted through the online Interpretation Submittal Form. The form is accessible at <http://go.asme.org/InterpretationRequest>. Upon submittal of the form, the Inquirer will receive an automatic e-mail confirming receipt.

If the Inquirer is unable to use the online form, he/she may mail the request to the Secretary of the B32 Standards Committee at the above address. The request for an interpretation should be clear and unambiguous. It is further recommended that the Inquirer submit his/her request in the following format:

Subject:	Cite the applicable paragraph number(s) and the topic of the inquiry in one or two words.
Edition:	Cite the applicable edition of the Standard for which the interpretation is being requested.
Question:	Phrase the question as a request for an interpretation of a specific requirement suitable for general understanding and use, not as a request for an approval of a proprietary design or situation. Please provide a condensed and precise question, composed in such a way that a "yes" or "no" reply is acceptable.
Proposed Reply(ies):	Provide a proposed reply(ies) in the form of "Yes" or "No," with explanation as needed. If entering replies to more than one question, please number the questions and replies.
Background Information:	Provide the Committee with any background information that will assist the Committee in understanding the inquiry. The Inquirer may also include any plans or drawings that are necessary to explain the question; however, they should not contain proprietary names or information.

Requests that are not in the format described above may be rewritten in the appropriate format by the Committee prior to being answered, which may inadvertently change the intent of the original request.

Moreover, ASME does not act as a consultant for specific engineering problems or for the general application or understanding of the Standard requirements. If, based on the inquiry information submitted, it is the opinion of the Committee that the Inquirer should seek assistance, the inquiry will be returned with the recommendation that such assistance be obtained.

ASME procedures provide for reconsideration of any interpretation when or if additional information that might affect an interpretation is available. Further, persons aggrieved by an interpretation may appeal to the cognizant ASME Committee or Subcommittee. ASME does not “approve,” “certify,” “rate,” or “endorse” any item, construction, proprietary device, or activity.

Attending Committee Meetings. The B32 Standards Committee regularly holds meetings and/or telephone conferences that are open to the public. Persons wishing to attend any meeting and/or telephone conference should contact the Secretary of the B32 Standards Committee. Future Committee meeting dates and locations can be found on the Committee Page at <http://go.asme.org/B32committee>.

Copyright ASME International
Provided by IHS Markit under license with ASME
No reproduction or networking permitted without license from IHS

.....

INTENTIONALLY LEFT BLANK

STAINLESS STEEL PIPE

1 SCOPE

This Standard covers the standardization of dimensions of welded and seamless wrought stainless steel pipe for high or low temperatures and pressures.

The word *pipe* is used, as distinguished from *tube*, to apply to tubular products of dimensions commonly used for pipeline and piping systems.

2 SIZE

The size of all pipe in Table 2-1 is identified by the dimensionless designator nominal pipe size (NPS) [diamètre nominal (DN)]. Pipes NPS 12 (DN 300) and smaller have outside diameters numerically larger than their corresponding sizes. In contrast, the outside diameters of tubes are numerically identical to the size number for all sizes.

The manufacture of pipe NPS $\frac{1}{8}$ (DN 6) to NPS 12 (DN 300), inclusive, is based on a standardized outside diameter (O.D.). This O.D. was originally selected so that pipe with a standard O.D. and having a wall thickness that was typical of the period would have an inside diameter (I.D.) approximately equal to the nominal size. Although there is no such relation between the existing standard thicknesses — O.D. and nominal size — these nominal sizes and standard O.D.s continue in use as “standard.”

The manufacture of pipe NPS 14 (DN 350) and larger proceeds on the basis of an O.D. corresponding to the nominal size.

3 MATERIALS

The dimensional standards for pipe described here are for products covered in ASTM specifications.

4 WALL THICKNESS

The nominal wall thicknesses are given in Table 2-1. The wall thicknesses for NPS 14 to NPS 22, inclusive (DN 350 to DN 550, inclusive), of Schedule 10S; NPS 12 (DN 300) of Schedule 40S; and NPS 10 and NPS 12 (DN 250 and DN 300) of Schedule 80S are not the same as those of ASME B36.10M, Welded and Seamless Wrought Steel Pipe. The suffix “S” in the schedule number is used to differentiate B36.19M pipe from B36.10M pipe. ASME B36.10M includes other pipe thicknesses that are also commercially available with stainless steel material.

5 WEIGHTS/MASSES

The nominal weights (masses)¹ of steel pipe are calculated values and are tabulated in Table 2-1.

(a) The nominal plain end weight, in pounds per foot, is calculated using the following formula:

$$W_{pe} = 10.69(D - t)t$$

where

D = outside diameter to the nearest 0.001 in. (the symbol D is used for O.D. only in mathematical equations or formulas)

t = specified wall thickness, rounded to the nearest 0.001 in.

W_{pe} = nominal plain end weight, rounded to the nearest 0.01 lb/ft

(b) The nominal plain end mass, in kilograms per meter, is calculated using the following formula:

$$M_{pe} = 0.0246615(D - t)t$$

where

D = outside diameter to the nearest 0.1 mm for outside diameters that are 16 in. (406.4 mm) and smaller, and 1.0 mm for outside diameters larger than 16 in. (406.4 mm) (the symbol D is used for O.D. only in mathematical equations or formulas)

M_{pe} = nominal plain end mass, rounded to the nearest 0.01 kg/m

t = specified wall thickness, rounded to the nearest 0.01 mm

6 PERMISSIBLE VARIATIONS

Variations in dimensions differ depending upon the method of manufacture employed in making the pipe to the various specifications available. Permissible variations for dimensions are indicated in each specification.

7 PIPE THREADS

Unless otherwise specified, the threads of threaded pipe shall conform to ASME B1.20.1, Pipe Threads, General Purpose (Inch).

Schedules 5S and 10S wall thicknesses do not permit threading in accordance with ASME B1.20.1.

¹ The different grades of stainless steel have different specific densities and hence their weights (masses) may be less than or greater than the values listed in Table 2-1 would indicate [see Table 2-1, General Note (e)].

8 WALL-THICKNESS SELECTION

When the selection of wall thickness depends primarily upon capacity to resist internal pressure under given conditions, the designer shall compute the exact value of wall thickness suitable for conditions for which the pipe is required, as prescribed in detail in the ASME

Boiler and Pressure Vessel Code, ASME B31 Code for Pressure Piping, or other similar code, whichever governs the construction. A thickness shall be selected from the schedules of nominal thickness contained in [Table 2-1](#) to suit the value computed to fulfill the conditions for which the pipe is desired.

Table 2-1 Dimensions of Welded and Seamless Stainless Steel Pipe and Nominal Weights (Masses) of Steel Pipe, Plain End

NPS (DN)	Schedule No.	Outside Diameter, in. (mm)	Wall Thickness, in. (mm)	Plain End Weight (Mass), lb/ft (kg/m)
$\frac{1}{8}$ (6)	5S	0.405 (10.3)	... [Note (1)]	...
$\frac{1}{8}$ (6)	10S	0.405 (10.3)	0.049 (1.24) [Note (1)]	0.19 (0.28)
$\frac{1}{8}$ (6)	40S	0.405 (10.3)	0.068 (1.73)	0.24 (0.37)
$\frac{1}{8}$ (6)	80S	0.405 (10.3)	0.095 (2.41)	0.31 (0.47)
$\frac{1}{4}$ (8)	5S	0.540 (13.7)	... [Note (1)]	...
$\frac{1}{4}$ (8)	10S	0.540 (13.7)	0.065 (1.65) [Note (1)]	0.33 (0.49)
$\frac{1}{4}$ (8)	40S	0.540 (13.7)	0.088 (2.24)	0.43 (0.63)
$\frac{1}{4}$ (8)	80S	0.540 (13.7)	0.119 (3.02)	0.54 (0.80)
$\frac{3}{8}$ (10)	5S	0.675 (17.1)	... [Note (1)]	...
$\frac{3}{8}$ (10)	10S	0.675 (17.1)	0.065 (1.65) [Note (1)]	0.42 (0.63)
$\frac{3}{8}$ (10)	40S	0.675 (17.1)	0.091 (2.31)	0.57 (0.84)
$\frac{3}{8}$ (10)	80S	0.675 (17.1)	0.126 (3.20)	0.74 (1.10)
$\frac{1}{2}$ (15)	5S	0.840 (21.3)	0.065 (1.65) [Note (1)]	0.54 (0.80)
$\frac{1}{2}$ (15)	10S	0.840 (21.3)	0.083 (2.11) [Note (1)]	0.67 (1.00)
$\frac{1}{2}$ (15)	40S	0.840 (21.3)	0.109 (2.77)	0.85 (1.27)
$\frac{1}{2}$ (15)	80S	0.840 (21.3)	0.147 (3.73)	1.09 (1.62)
$\frac{3}{4}$ (20)	5S	1.050 (26.7)	0.065 (1.65) [Note (1)]	0.68 (1.02)
$\frac{3}{4}$ (20)	10S	1.050 (26.7)	0.083 (2.11) [Note (1)]	0.86 (1.28)
$\frac{3}{4}$ (20)	40S	1.050 (26.7)	0.113 (2.87)	1.13 (1.69)
$\frac{3}{4}$ (20)	80S	1.050 (26.7)	0.154 (3.91)	1.48 (2.20)
1 (25)	5S	1.315 (33.4)	0.065 (1.65) [Note (1)]	0.87 (1.29)
1 (25)	10S	1.315 (33.4)	0.109 (2.77) [Note (1)]	1.41 (2.09)
1 (25)	40S	1.315 (33.4)	0.133 (3.38)	1.68 (2.50)
1 (25)	80S	1.315 (33.4)	0.179 (4.55)	2.17 (3.24)
$1\frac{1}{4}$ (32)	5S	1.660 (42.2)	0.065 (1.65) [Note (1)]	1.11 (1.65)
$1\frac{1}{4}$ (32)	10S	1.660 (42.2)	0.109 (2.77) [Note (1)]	1.81 (2.69)
$1\frac{1}{4}$ (32)	40S	1.660 (42.2)	0.140 (3.56)	2.27 (3.39)
$1\frac{1}{4}$ (32)	80S	1.660 (42.2)	0.191 (4.85)	3.00 (4.47)
$1\frac{1}{2}$ (40)	5S	1.900 (48.3)	0.065 (1.65) [Note (1)]	1.28 (1.90)
$1\frac{1}{2}$ (40)	10S	1.900 (48.3)	0.109 (2.77) [Note (1)]	2.09 (3.11)
$1\frac{1}{2}$ (40)	40S	1.900 (48.3)	0.145 (3.68)	2.72 (4.05)
$1\frac{1}{2}$ (40)	80S	1.900 (48.3)	0.200 (5.08)	3.63 (5.41)
2 (50)	5S	2.375 (60.3)	0.065 (1.65) [Note (1)]	1.61 (2.39)
2 (50)	10S	2.375 (60.3)	0.109 (2.77) [Note (1)]	2.64 (3.93)
2 (50)	40S	2.375 (60.3)	0.154 (3.91)	3.66 (5.44)
2 (50)	80S	2.375 (60.3)	0.218 (5.54)	5.03 (7.48)
$2\frac{1}{2}$ (65)	5S	2.875 (73)	0.083 (2.11) [Note (1)]	2.48 (3.69)
$2\frac{1}{2}$ (65)	10S	2.875 (73)	0.120 (3.05) [Note (1)]	3.53 (5.26)

Table 2-1 Dimensions of Welded and Seamless Stainless Steel Pipe and Nominal Weights (Masses) of Steel Pipe, Plain End (Cont'd)

NPS (DN)	Schedule No.	Outside Diameter, in. (mm)	Wall Thickness, in. (mm)	Plain End Weight (Mass), lb/ft (kg/m)
2½ (65)	40S	2.875 (73)	0.203 (5.16)	5.80 (8.63)
2½ (65)	80S	2.875 (73)	0.276 (7.01)	7.67 (11.41)
3 (80)	5S	3.500 (88.9)	0.083 (2.11) [Note (1)]	3.03 (4.52)
3 (80)	10S	3.500 (88.9)	0.120 (3.05) [Note (1)]	4.34 (6.46)
3 (80)	40S	3.500 (88.9)	0.216 (5.49)	7.58 (11.29)
3 (80)	80S	3.500 (88.9)	0.300 (7.62)	10.26 (15.27)
3½ (90)	5S	4.000 (101.6)	0.083 (2.11) [Note (1)]	3.48 (5.18)
3½ (90)	10S	4.000 (101.6)	0.120 (3.05) [Note (1)]	4.98 (7.41)
3½ (90)	40S	4.000 (101.6)	0.226 (5.74)	9.12 (13.57)
3½ (90)	80S	4.000 (101.6)	0.318 (8.08)	12.52 (18.64)
4 (100)	5S	4.500 (114.3)	0.083 (2.11) [Note (1)]	3.92 (5.84)
4 (100)	10S	4.500 (114.3)	0.120 (3.05) [Note (1)]	5.62 (8.37)
4 (100)	40S	4.500 (114.3)	0.237 (6.02)	10.80 (16.08)
4 (100)	80S	4.500 (114.3)	0.337 (8.56)	15.00 (22.32)
5 (125)	5S	5.563 (141.3)	0.109 (2.77) [Note (1)]	6.36 (9.46)
5 (125)	10S	5.563 (141.3)	0.134 (3.40) [Note (1)]	7.78 (11.56)
5 (125)	40S	5.563 (141.3)	0.258 (6.55)	14.63 (21.77)
5 (125)	80S	5.563 (141.3)	0.375 (9.53)	20.80 (30.97)
6 (150)	5S	6.625 (168.3)	0.109 (2.77) [Note (1)]	7.59 (11.31)
6 (150)	10S	6.625 (168.3)	0.134 (3.40) [Note (1)]	9.30 (13.83)
6 (150)	40S	6.625 (168.3)	0.280 (7.11)	18.99 (28.26)
6 (150)	80S	6.625 (168.3)	0.432 (10.97)	28.60 (42.56)
8 (200)	5S	8.625 (219.1)	0.109 (2.77) [Note (1)]	9.92 (14.78)
8 (200)	10S	8.625 (219.1)	0.148 (3.76) [Note (1)]	13.41 (19.97)
8 (200)	40S	8.625 (219.1)	0.322 (8.18)	28.58 (42.55)
8 (200)	80S	8.625 (219.1)	0.500 (12.70)	43.43 (64.64)
10 (250)	5S	10.750 (273.0)	0.134 (3.40) [Note (1)]	15.21 (22.61)
10 (250)	10S	10.750 (273.0)	0.165 (4.19) [Note (1)]	18.67 (27.78)
10 (250)	40S	10.750 (273.0)	0.365 (9.27)	40.52 (60.29)
10 (250)	80S	10.750 (273.0)	0.500 (12.70) [Note (2)]	54.79 (81.53)
12 (300)	5S	12.750 (323.8)	0.156 (3.96) [Note (1)]	21.00 (31.24)
12 (300)	10S	12.750 (323.8)	0.180 (4.57) [Note (1)]	24.19 (35.98)
12 (300)	40S	12.750 (323.8)	0.375 (9.53) [Note (2)]	49.61 (73.86)
12 (300)	80S	12.750 (323.8)	0.500 (12.70) [Note (2)]	65.48 (97.44)
14 (350)	5S	14.000 (355.6)	0.156 (3.96) [Note (1)]	23.09 (34.34)
14 (350)	10S	14.000 (355.6)	0.188 (4.78) [Notes (1), (2)]	27.76 (41.36)
14 (350)	40S	14.000 (355.6)	0.375 (9.53) [Note (2)]	54.62 (81.33)

Table 2-1 Dimensions of Welded and Seamless Stainless Steel Pipe and Nominal Weights (Masses) of Steel Pipe, Plain End (Cont'd)

NPS (DN)	Schedule No.	Outside Diameter, in. (mm)	Wall Thickness, in. (mm)	Plain End Weight (Mass), lb/ft (kg/m)
14 (350)	80S	14.000 (355.6)	0.500 (12.70) [Note (2)]	72.16 (107.40)
16 (400)	5S	16.000 (406.4)	0.165 (4.19) [Note (1)]	27.93 (41.56)
16 (400)	10S	16.000 (406.4)	0.188 (4.78) [Notes (1), (2)]	31.78 (47.34)
16 (400)	40S	16.000 (406.4)	0.375 (9.53) [Note (2)]	62.64 (93.27)
16 (400)	80S	16.000 (406.4)	0.500 (12.70) [Note (2)]	82.85 (123.31)
18 (450)	5S	18.000 (457)	0.165 (4.19) [Note (1)]	31.46 (46.79)
18 (450)	10S	18.000 (457)	0.188 (4.78) [Notes (1), (2)]	35.80 (53.31)
18 (450)	40S	18.000 (457)	0.375 (9.53) [Note (2)]	70.65 (...)
18 (450)	80S	18.000 (457)	0.500 (12.70) [Note (2)]	93.54 (...)
20 (500)	5S	20.000 (508)	0.188 (4.78) [Note (1)]	39.82 (59.32)
20 (500)	10S	20.000 (508)	0.218 (5.54) [Notes (1), (2)]	46.10 (68.65)
20 (500)	40S	20.000 (508)	0.375 (9.53) [Note (2)]	78.67 (117.15)
20 (500)	80S	20.000 (508)	0.500 (12.70) [Note (2)]	104.23 (155.13)
22 (550)	5S	22.000 (559)	0.188 (4.78) [Note (1)]	43.84 (65.33)
22 (550)	10S	22.000 (559)	0.218 (5.54) [Notes (1), (2)]	50.76 (75.62)
22 (550)	40S	22.000 (559)	...	...
22 (550)	80S	22.000 (559)	...	...
24 (600)	5S	24.000 (610)	0.218 (5.54) [Note (1)]	55.42 (82.58)
24 (600)	10S	24.000 (610)	0.250 (6.35) [Note (1)]	63.47 (94.53)
24 (600)	40S	24.000 (610)	0.375 (9.53) [Note (2)]	94.71 (141.12)
24 (600)	80S	24.000 (610)	0.500 (12.70) [Note (2)]	125.61 (187.07)
30 (750)	5S	30.000 (762)	0.250 (6.35) [Note (1)]	79.51 (118.34)
30 (750)	10S	30.000 (762)	0.312 (7.92) [Note (1)]	99.02 (147.29)
30 (750)	40S	30.000 (762)	...	...
30 (750)	80S	30.000 (762)	...	...

GENERAL NOTES:

- (a) 1 in. = 25.4 mm.
 (b) For tolerances, see [section 6](#).
 (c) 1 lb/ft = 1.4895 kg/m.
 (d) Weights (masses) are given in pounds per linear foot (kilograms per meter) and are for carbon steel pipe with plain ends.
 (e) The different grades of stainless steel permit considerable variations in weight (mass). The ferritic stainless steels may be about 5% less, and the austenitic stainless steels about 2% greater, than the values shown in this Table, which are based on weights (masses) for carbon steel.

NOTES:

- (1) These wall thicknesses do not permit threading in accordance with ASME B1.20.1.
 (2) These dimensions do not conform to ASME B36.10M.

INTENTIONALLY LEFT BLANK

B32/B36 AMERICAN NATIONAL STANDARDS FOR PRODUCT SIZES

B32.5-1977 (R2010)	Preferred Metric Sizes for Tubular Metal Products Other Than Pipe
B32.100-2016	Preferred Metric Sizes for Flat, Round, Square, Rectangular, and Hexagonal Metal Products
B36.10M-2018	Welded and Seamless Wrought Steel Pipe
B36.19M-2018	Stainless Steel Pipe

The ASME Publications Catalog shows a complete list of all the Standards published by the Society. For a complimentary catalog, or the latest information about our publications, call 1-800-THE-ASME (1-800-843-2763).

ASME Services

ASME is committed to developing and delivering technical information. At ASME's Customer Care, we make every effort to answer your questions and expedite your orders. Our representatives are ready to assist you in the following areas:

ASME Press	Member Services & Benefits	Public Information
<i>Codes & Standards</i>	Other ASME Programs	Self-Study Courses
Credit Card Orders	Payment Inquiries	Shipping Information
IMechE Publications	Professional Development	Subscriptions/Journals/Magazines
Meetings & Conferences	Short Courses	Symposia Volumes
Member Dues Status	Publications	Technical Papers

How can you reach us? It's easier than ever!

There are four options for making inquiries* or placing orders. Simply mail, phone, fax, or E-mail us and a Customer Care representative will handle your request.

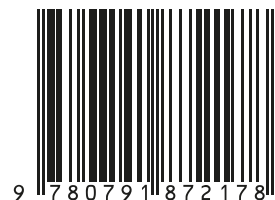
<i>Mail</i>	<i>Call Toll Free</i>	<i>Fax—24 hours</i>	<i>E-Mail—24 hours</i>
ASME	US & Canada: 800-THE-ASME	973-882-1717	customercare@asme.org
150 Clove Road, 6th Floor	(800-843-2763)	973-882-5155	
Little Falls, New Jersey	Mexico: 95-800-THE-ASME		
07424-2139	(95-800-843-2763)		

*Customer Care staff are not permitted to answer inquiries about the technical content of this code or standard. Information as to whether or not technical inquiries are issued to this code or standard is shown on the copyright page. All technical inquiries must be submitted in writing to the staff secretary. Additional procedures for inquiries may be listed within.

ASME B36.19M-2018

Copyright ASME International
Provided by IHS Markit under license with ASME
No reproduction or networking permitted without license from IHS

I S B N 978-0-7918-7217-8



9 780791 872178



M 0 1 3 1 8